Project Design Phase-I
Solution Architecture

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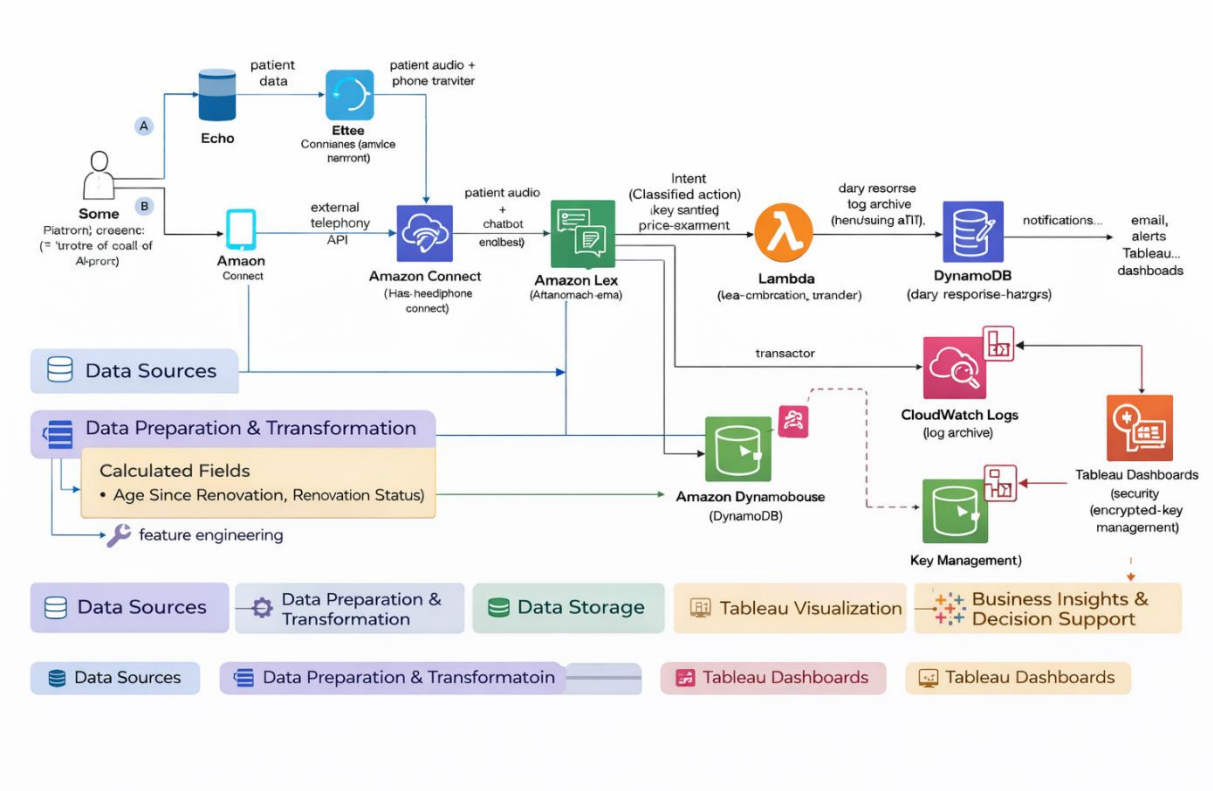
Project Name	Project - **Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau**
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture - Visualizing Housing Market Trends using Tableau:



Solution Architecture and Data Flow - Visualizing Housing Market Trends using Tableau

Our Project's Solution Architecture:

The solution architecture for this project is structured across five key layers, each playing a critical role in transforming raw housing data into interactive, actionable Tableau dashboards.

Layer 1: Data Source Layer

The foundation of the architecture is the raw housing dataset — Transformed_Housing_Data2.csv — containing 21,609 property records. Each record includes key variables such as sale price, number of bedrooms, bathrooms, floors, basement area in sqft, house age in years, and renovation status. This dataset serves as the single source of truth for all visualizations and analysis performed within the solution.

Layer 2: Data Processing and Transformation Layer

Before loading into Tableau, the raw dataset undergoes data cleaning and transformation. This includes handling missing values, standardizing column formats, creating calculated fields such as Sale_Price (bin) for price range grouping, deriving the Age of House in years, and creating a binary Ever Renovated field. These transformations ensure the data is structured, consistent, and ready for visual analysis.

Layer 3: Visualization Layer (Tableau Dashboard)

The processed data is connected directly to Tableau Desktop where the following visualizations are built and assembled into interactive dashboards:

- **KPI Summary Cards:** Displaying average sale price ($511,619), total basement area (38.6M sqft), and total property count (21,609).
- **Bubble Chart:** Market concentration across housing price ranges from $100K to $1.1M, sized by count and colored by average sale price.
- **Bar Chart:** Distribution of house sale prices showing a steady decline from the $250K–$400K peak down to $50K and $1.1M segments.
- **Grouped Bar Chart:** House age distribution by number of bathrooms, bedrooms, and floors across age groups ranging from 4 to 15 years.
- **Pie Chart:** Distribution of house age by renovation status among 204 renovated properties, broken down across 10 age groups.

Layer 4: Presentation Layer (Story and Dashboard)

All individual visualizations are organized into two comprehensive Tableau dashboards — a Comprehensive House Data Analysis dashboard and a Story dashboard with four narrative slides. Each story slide presents a key insight: house age influenced by bedrooms, renovation status overview, mid-range price market dominance, and the inverse relationship between sale price and number of houses sold.

Layer 5: End User Layer (Consumers)

The final output is consumed by real estate agents, property investors, home buyers, and housing market analysts who interact with the dashboards to explore market trends, compare property features, and make data-driven pricing and investment decisions. The dashboards are designed to be intuitive and accessible to non-technical users without requiring any coding or data expertise.

Architecture Data Flow:

CSV Dataset (21,609 Records) → Data Cleaning & Transformation → Tableau Data Connection → Calculated Fields & Binning → Individual Worksheets → Dashboard Assembly → Story Creation → End User Insights